

MUKDAHAN, Thailand

WMO: 483830

Lat: 16.533N

Lon: 104.717E

Elev: 140

StdP: 99.65

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	14.0	15.2	8.6	7.0	17.2	9.9	7.7	18.1	6.2	23.2	5.3	24.2	0.9	50	0.302

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.0	37.9	25.3	36.7	25.2	35.6	25.2	27.3	32.7	27.1	32.4	26.8	31.9	1.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.2	21.9	29.3	25.8	21.5	29.1	25.6	21.2	29.0	87.8	32.6	86.4	32.5	85.4	32.2	33.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
4.7	3.8	3.3	DB	11.8	39.6	1.5	1.1	10.7	40.3	9.7	40.9	8.9	41.5	7.8	42.3	
			WB	9.9	28.5	1.5	1.2	8.8	29.4	8.0	30.0	7.2	30.7	6.1	31.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.3	23.6	25.7	28.5	30.2	29.8	29.2	28.3	28.1	28.0	27.4	25.6	23.5
	DBStd	3.10	2.66	3.34	3.13	2.42	1.88	1.55	1.61	1.34	1.31	1.55	2.17	2.64
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	3	1	1	0	0	0	0	0	0	0	0	0	1
	CDD10.0	6323	421	441	574	606	613	576	568	560	540	539	468	419
	CDD18.3	3285	163	208	316	356	354	326	309	302	290	281	218	162
	CDH23.3	35563	1438	2141	3792	4523	4334	3860	3352	3154	2976	2768	1926	1299
CDH26.7	15541	563	1028	2018	2466	2108	1683	1256	1083	1060	1077	739	460	
Wind	WSAvg	0.8	1.1	1.0	1.0	0.7	0.6	0.6	0.6	0.6	0.5	0.8	1.1	1.3
Precipitation	PrecAvg	1564	3	17	34	83	202	282	259	341	272	69	5	2
	PrecMax	2095	40	73	74	216	337	588	406	611	514	234	26	17
	PrecMin	1158	0	0	0	17	101	137	63	196	94	15	0	0
	PrecStd	260	10	25	26	46	71	109	94	129	126	57	7	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.7	37.2	38.9	39.8	38.5	36.2	35.5	34.2	34.2	34.0	34.6	33.7
		MCWB	23.2	23.4	24.4	25.3	26.1	26.5	26.2	26.9	26.7	26.0	25.3	24.2
	2%	DB	32.9	35.6	37.4	38.2	36.7	34.8	33.8	33.2	33.2	33.1	33.3	32.3
		MCWB	22.7	23.5	24.5	25.6	26.1	26.5	26.4	26.7	26.3	25.5	24.8	23.4
	5%	DB	31.5	34.1	36.2	36.9	35.3	33.8	32.8	32.2	32.4	32.4	32.0	31.0
		MCWB	22.3	23.3	24.4	25.7	26.1	26.4	26.3	26.5	26.1	25.2	24.4	22.7
	10%	DB	30.0	32.5	34.9	35.5	33.9	32.8	31.8	31.3	31.5	31.6	30.7	29.4
		MCWB	21.6	23.0	24.4	25.6	26.1	26.4	26.2	26.2	25.9	24.8	23.7	21.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.8	25.5	26.6	27.5	27.7	27.6	27.4	27.5	27.5	27.0	26.3	25.1
		MCDB	31.5	33.7	34.8	35.5	33.7	33.0	32.0	32.2	32.3	32.4	32.4	32.0
	2%	WB	23.8	24.8	26.0	26.8	27.2	27.2	27.0	27.1	27.0	26.4	25.7	24.2
		MCDB	30.3	32.8	34.3	34.7	33.1	32.3	31.7	31.5	31.6	31.5	31.3	30.7
	5%	WB	23.0	24.2	25.5	26.4	26.8	26.9	26.6	26.7	26.6	25.9	25.0	23.4
		MCDB	30.0	31.6	33.4	34.1	32.5	31.9	31.1	30.8	30.8	30.7	30.4	29.5
	10%	WB	22.2	23.6	25.0	26.0	26.5	26.6	26.3	26.5	26.3	25.5	24.3	22.5
		MCDB	28.7	30.4	32.6	33.5	32.0	31.3	30.5	30.2	30.0	29.9	29.3	28.3
Mean Daily Temperature Range	5% DB	MDBR	12.4	12.4	11.3	10.0	8.2	6.9	6.2	6.1	6.6	8.0	9.7	10.8
		MCDBR	13.6	13.3	12.2	11.1	9.8	8.2	7.8	7.6	8.0	9.0	10.8	12.1
		MCWBR	4.8	3.9	3.3	3.0	2.5	2.3	2.4	2.5	2.7	3.1	3.8	4.5
	5% WB	MCDBR	12.4	11.8	11.1	10.0	8.4	7.4	7.1	7.0	7.1	8.0	9.7	11.2
MCWBR		4.8	4.0	3.5	3.2	2.8	2.6	2.6	2.6	2.6	3.1	3.6	4.5	
Clear-Sky Solar Irradiance	taub		0.459	0.570	0.659	0.621	0.510	0.474	0.464	0.469	0.504	0.512	0.459	0.456
	taud		2.154	1.859	1.694	1.821	2.149	2.273	2.300	2.274	2.165	2.113	2.238	2.187
	Ebn at Noon		825	748	697	727	801	822	832	835	808	789	820	814
	Edh at Noon		145	204	248	219	156	136	133	137	153	156	133	138
All-Sky Solar Radiation	RadAvg		4.97	5.26	5.20	5.51	5.34	4.95	4.55	4.43	4.58	4.98	4.84	4.70
	RadStd		0.26	0.32	0.18	0.30	0.30	0.34	0.47	0.32	0.42	0.40	0.19	0.17

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
Station Only	+0.36	N/A	N/A	N/A	N/A	+0.14	N/A	N/A	N/A	N/A
Regional (13 neighbors)	+0.27	N/A	N/A	N/A	+0.16	+0.15	N/A	N/A	N/A	N/A

Nomenclature: See separate page